Assignment 4

Tuesday, September 30, 2025 10:13 PM

CHE 312

Q1. (a) $q'' = k_{Al} \frac{\Delta T}{L}$

$L = 400\ \mu m = 4 \times 10^{-4}\ m$

$q''_x = 237 \cdot \frac{20}{4 \times 10^{-4}} = 1.185 \times 10^{7}\ W m^{-2}$

$L = 400\ nm = 4 \times 10^{-7}\ m$

$q''_x = 237 \cdot \frac{20}{4 \times 10^{-4}} = 1.185 \times 10^{10}\ W m^{-2}$

(b)

$L = 400\ \mu m$

$q''_x = 0.0263 \times \frac{20}{4 \times 10^{-4}} = 1.315 \times 10^{3}\ W m^{-2}$

$L = 400\ nm$

$q''_x = 0.0263 \times \frac{20}{4 \times 10^{-7}} = 1.315 \times 10^{6}\ W m^{-2}$

(c) Aluminium - Air - Aluminium

· $L_{tot} = 400\ \mu m$, $\delta = 40\ \mu m$

$L_{air} = 400 - 2 \times 40 = 320\ \mu m$

$R_{tot} = \frac{320 \times 10^{-6}}{0.0263} + 2 \frac{4 \times 10^{-5}}{237} = 0.0121\ m^2 K W^{-1}$

$q''_x = \frac{20}{0.0121} = 1.644 \times 10^{3}\ W m^{-2}$

· $L_{tot} = 400\ nm$, $\delta = 40\ nm$

$L_{air} = 320$

(similar to previous layer)

$\therefore q = 1.644 \times 10^{4}\ W m^{-2}$

Q2) $Q = -k A(r) \frac{dT}{dr}$, $A(r) = 4\pi r^2$

$Q = \frac{4\pi k (T_1 - T_2)}{\left(\frac{1}{r_1} - \frac{1}{r_2}\right)} \Rightarrow k = \frac{Q}{4\pi(T_1 - T_2)}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)$

$k = \frac{60}{4 \times 3.14 \times 150}\left(\frac{1}{0.10} - \frac{1}{0.15}\right) = \underline{0.0106\ W m^{-1} K^{-1}}$

Q3)

Heat flow normal to layers : series resistances

$k_\perp = \frac{1}{\frac{1}{2}\left(\frac{1}{k_c} + \frac{1}{k_g}\right)} = \frac{2}{\frac{1}{k_c} + \frac{1}{k_g}} = \frac{2 k_c k_g}{k_c + k_g}$

Heat flow || to layers : || paths (area fraction 50/50)

$k_{\parallel} = \frac{1}{2}(k_c + k_g)$

$\frac{Q_{\parallel}}{Q_\perp} = \frac{k_{\parallel}}{k_\perp} = \frac{\frac{1}{2}(k_c + k_g)}{\frac{2 k_c k_g}{k_c + k_g}} = \frac{(k_c + k_g)^2}{4 k_c k_g} = \frac{1}{4}\left(\frac{k_c}{k_g} + \frac{k_g}{k_c} + 2\right)$

Implication: $\frac{Q_{\parallel}}{Q_\perp} \geq 1$ with equality only if $k_c = k_g$, $\therefore$ cake always conducts better when heat flows parallel to the layers.